

# 2501-PYL1003 (Mechanics & STR) — Practice Mid-Sem Paper 1

Full Marks: 50 | Time: 2 hours | Instructor Style: Pradipta Ghosh (IIT Delhi)

## Instructions

An MCQ can also be an MSQ (multiple correct options). No credit without justification/derivation. Show all intermediate steps. Numerical answers without units lose marks. Approximations must be justified explicitly.

**Q1.** A physicist proposes an empirical formula for the drag force on a sphere moving through a fluid:

$$F = k \rho^a v^b r^c \eta^d$$

where  $\rho$  is fluid density,  $v$  is speed,  $r$  is the sphere's radius, and  $\eta$  is the fluid's coefficient of viscosity (dimension  $ML^{-1}T^{-1}$ ), and  $k$  is a dimensionless constant.

(i) Using dimensional analysis alone, show that the exponents cannot be uniquely determined by dimensional analysis alone if all four of  $a, b, c, d$  are treated as independent unknowns. State precisely how many independent constraints dimensional homogeneity provides, and how many free parameters remain. **(3)**

(ii) It is separately known (from experiment) that  $F$  is directly proportional to  $\eta$  and independent of  $\rho$  at low speeds. Using this additional physical input together with dimensional analysis, determine  $a, b, c, d$  uniquely. **(3)**

**Q2 (MSQ).** A quantity  $X$  is defined as  $X = \frac{(\text{Planck's constant}) \times (\text{speed of light})}{(\text{Gravitational constant})}$ . Which of the following statement(s) is/are correct? **(1+1+1 = 3)**

a.  $X$  has the dimension of mass squared. b.  $X$  has the dimension of  $M^2 L^0 T^0$ . c.  $\sqrt{X}$  has the dimension of the "Planck mass." d.  $X$  is dimensionless.

Justify your choice(s) by explicitly working out the dimensional formula.

**Q3.** A particle moves along a straight line such that its acceleration is given by  $a = -kx^3$ , where  $x$  is its displacement from the origin and  $k > 0$  is a constant. At  $t = 0$ , the particle is at  $x = x_0$  and momentarily at rest.

(i) Using  $a = v \frac{dv}{dx}$ , find  $v$  as an explicit function of  $x$ . **(3)**

(ii) Show that the particle executes oscillatory motion, and determine the two turning points of the motion. **(2)**

(iii) Set up (but you need **not** evaluate in closed form) the exact integral expression for the time period of this oscillation in terms of  $k$ ,  $x_0$ , and  $x$ . **(2)**

**Q4.** A leaking bucket of initial mass  $M_0$  (including water) is pulled vertically upward from rest by a constant force  $F > M_0 g$ . Water leaks out at a constant rate  $\lambda$  (kg/s), so the mass at time  $t$  is  $M(t) = M_0 - \lambda t$ . The leaking water carries away no vertical momentum relative to the bucket (i.e., it just drips out with the bucket's instantaneous velocity).

(i) Write down the correct equation of motion for the bucket, explaining carefully why this is **not** a variable-mass rocket-thrust problem despite the mass changing with time. **(2)**

(ii) Solve for  $v(t)$ , the velocity of the bucket, valid until the bucket empties at  $t = M_0/\lambda$ . **(3)**

**Q5.** A block of mass  $m$  rests on a frictionless wedge of mass  $M$  and incline angle  $\theta$ , which itself sits on a frictionless horizontal floor. The wedge is free to slide horizontally. The system starts at rest.

(i) Using the constraint that the block always remains on the wedge's surface, find the horizontal acceleration of the wedge in terms of  $m$ ,  $M$ ,  $\theta$ ,  $g$ . **(4)**

(ii) Find the normal force between the block and the wedge. Verify that your expression correctly reduces to  $mg \cos \theta$  in the limit  $M \rightarrow \infty$ . **(3)**

**Q6.** The position of a point  $P$  in Cartesian coordinates is  $(x, y, z) = (-6, 8, 15)$ .

(i) Find the cylindrical polar coordinates  $(\rho, \phi, z)$  of  $P$ , taking care to place  $\phi$  in the correct quadrant. **(2)**

(ii) Find the spherical polar coordinates  $(r, \theta, \phi)$  of  $P$ . **(2)**

(iii) A second point  $Q$  has spherical coordinates  $r = 17$ ,  $\theta = 61.93^\circ$ ,  $\phi = 126.87^\circ$  (rounded to two decimals). Determine whether  $Q$  and  $P$  are the same point. Show your work. **(2)**

**Q7.** Starting from  $\vec{r} = r \sin \theta \cos \phi \hat{i} + r \sin \theta \sin \phi \hat{j} + r \cos \theta \hat{k}$  and the definition  $\hat{u} = \frac{\partial \vec{r} / \partial u}{|\partial \vec{r} / \partial u|}$  for  $u = r, \theta, \phi$ :

(i) Derive the expression for  $\hat{\theta}$  in terms of  $\hat{i}, \hat{j}, \hat{k}$ . **(3)**

(ii) Hence **prove**, without simply quoting the standard result, that

$$\frac{d\hat{r}}{dt} = \dot{\theta} \hat{\theta} + \dot{\phi} \sin \theta \hat{\phi}$$

by direct differentiation of your Cartesian expression for  $\hat{r}$  with respect to time. **(4)**

**Q8.** A vector field is given in spherical polar coordinates as  $\vec{A} = 2r \hat{r} - r \sin \theta \hat{\theta} + 4 \hat{\phi}$ .

(i) At the point  $r = 2$ ,  $\theta = 60^\circ$ ,  $\phi = 30^\circ$ , express  $\hat{r}$ ,  $\hat{\theta}$ ,  $\hat{\phi}$  in terms of  $\hat{i}$ ,  $\hat{j}$ ,  $\hat{k}$  (two decimal places). **(3)**

(ii) Hence compute the Cartesian components of  $\vec{A}$  at this point. **(2)**

**Q9 (MSQ).** For a particle undergoing motion under an inverse-square central force, the orbit equation is  $r = \frac{p}{1 + \epsilon \cos \theta}$ . Which of the following is/are true? **(1+1+1+1 = 4)**

a. For  $\epsilon = 0$ , the orbit is a circle and speed is constant everywhere on the orbit. b. For  $0 < \epsilon < 1$ , the orbit is an ellipse with the force center at one focus, and speed is maximum at  $\theta = 0$ . c. The semi-latus rectum  $p$  is related to the angular momentum  $L$  and reduced mass  $\mu$  via  $p = L^2 / (\mu k)$  for a force  $F = -k/r^2$ , independent of the total energy  $E$ . d. For  $\epsilon > 1$ , the orbit is unbound and the particle's speed at  $r \rightarrow \infty$  is zero.

**Q10.** A comet orbits the Sun on a highly eccentric ellipse. At perihelion, its distance from the Sun is  $r_p$  and its speed is  $v_p$ ; at aphelion, its distance is  $r_a$  and speed  $v_a$ . It is observed that  $r_a = 4 r_p$ .

(i) Using conservation of angular momentum, find the ratio  $v_p/v_a$ . **(2)**

(ii) Using conservation of energy together with your result from (i), derive the orbit's eccentricity  $\epsilon$  purely in terms of the given ratio  $r_a/r_p$  (do **not** assume the perihelion-aphelion speed ratio result from a textbook — derive it from  $E$  and  $L$  conservation). **(4)**

**Q11.** In the pulley system shown conceptually below, a single light, inextensible string is threaded as follows: one end is fixed to the ceiling, it passes under a movable pulley  $P_1$  (from which mass  $m_1$  hangs), over a fixed pulley attached to the ceiling, and then under a second movable pulley  $P_2$  (from which mass  $m_2$  hangs), with its other end held by a person pulling with tension  $T_0$  directly. All pulleys are massless and frictionless, and the string is massless.

(i) Write down the constraint relating the accelerations of  $m_1$ ,  $m_2$ , and the point where the person's hand moves, given that the string is inextensible. **(3)**

(ii) If the person pulls such that  $T_0 = 20$  N, and  $m_1 = m_2 = 0.5$  kg, find the acceleration of each mass (magnitude and direction), taking  $g = 10$  m/s<sup>2</sup>. **(4)**

(iii) Draw a properly labeled, scaled free-body diagram for  $m_1$  and for  $m_2$ . **(2)**

**Q12.** A particle of mass 2 kg moves along the  $x$ -axis under a force  $F(x) = -\frac{18}{x^3}$  N (with  $x$  in meters,  $x > 0$ ). At  $x = 3$  m, the particle has speed 2 m/s moving in the  $+x$  direction.

(i) Using the work-energy theorem, find the speed of the particle when it reaches  $x = 1$  m. **(3)**

(ii) Determine whether the particle can ever reach  $x = 0$ , justifying your answer using energy conservation (i.e., identify the classically forbidden region, if any). **(2)**

**Q13.** Two blocks of mass  $m$  and  $2m$  are connected by a light spring of natural length  $L_0$  and spring constant  $k$ , resting on a frictionless horizontal surface. The blocks are pulled apart by an additional length  $d$  and released from rest simultaneously.

(i) Using the center-of-mass frame, find the amplitude of oscillation of **each** block about its own equilibrium position, in terms of  $d$ . **(4)**

(ii) Find the angular frequency of the relative oscillation between the two blocks in terms of  $m$  and  $k$ . **(2)**

**Q14.** Prove, from first principles (i.e., starting from Newton's second law for a system of particles), that the total torque about a **fixed** point  $O$  equals the rate of change of angular momentum about that same point, for a system of  $N$  particles interacting via internal forces that obey Newton's third law **and** act along the line joining each pair of particles. Clearly state where the "central forces" assumption is used in the proof. **(5)**

**Q15.** A block of mass 5 kg is placed on a rough horizontal surface. The coefficient of static friction is  $\mu_s = 0.4$  and kinetic friction is  $\mu_k = 0.3$ . A horizontal force  $F(t) = 3t$  N ( $t$  in seconds) is applied starting at  $t = 0$ .

(i) Find the time  $t_1$  at which the block just begins to move. **(2)**

(ii) For  $t > t_1$ , write the differential equation of motion for the block's velocity, and solve for  $v(t)$  until  $t = 2t_1$ . Take  $g = 10 \text{ m/s}^2$ . **(4)**

(End of Paper 1 — 50 marks)